Zapiski nauchnykh
seminarov POMI
Vol. , 2026

Valentin Malykh, Dmitrii Zelenin

VORTEX-HRM: HIERARCHICAL RETRIEVAL-AUGMENTED GENERATION FOR MULTI-HOP QUESTION ANSWERING

ABSTRACT. We present VORTEX-HRM, an agentic retrieval-augmented generation system for multi-hop question answering. Unlike prior work that relies on LLM-driven tool selection or flat ReAct loops, VORTEX separates planning from execution using a fact-free planner and a fixed chained retrieval pipeline. The Gravitational Core (planner) holds zero factual knowledge in its weights—it is a pure structural router communicating via an XML-based protocol. The Centrifugal Ingestor (executor) resolves sub-questions through a deterministic pipeline: keyword search, chunk reading, and LLM condensation. The cycle repeats—each rotation is a spiral—until convergence via entropic collapse. On a 50-question multi-domain benchmark with three LLMs, VORTEX achieves 74% Contains accuracy (qwen2.5:7b, T4 GPU), outperforming Naive RAG by +6%. The architecture reduces hallucination by 2.4× (10% vs 24% no-evidence rate). Ablation confirms spirals are critical: single-pass yields 60%, five spirals reach 74%, plateauing thereafter. However, VORTEX is sensitive to LLM capability—on weaker models accuracy drops below baseline. We further introduce a Fast/Slow Router that dynamically gates between Naive RAG and VORTEX using the LLM's own uncertainty signal ("Insufficient evidence."). The router achieves 82%, outperforming both naive RAG (+14%) and VORTEX alone (+8%) with zero additional training. The system is offline-first, config-driven, and validated across three backends (mock, Ollama, OpenAI).

§1. INTRODUCTION

Multi-hop question answering requires retrieving and reasoning over multiple documents. Standard RAG [1] retrieves once and answers in a single step—it cannot chain evidence across documents.

Recent agentic approaches attempt to address this. A-RAG [2] augments RAG with LLM-driven tool selection, but this wastes tokens on routing decisions and introduces failure modes. ReAct [3] interleaves reasoning and acting but lacks a convergence signal—it terminates by hard step limit or the LLM's arbitrary decision. RAPTOR [4] builds hierarchical document trees via recursive summarization, requiring expensive index-time preprocessing.

Key words and phrases: retrieval-augmented generation and multi-hop QA and agentic systems and hierarchical retrieval.

We introduce VORTEX, a system that replaces LLM tool selection with a fixed chained pipeline, replaces arbitrary termination with entropic collapse, and replaces parametric knowledge with fact-free routing. The system is offline-first, config-driven, and validated on CPU hardware.

§2. Related Work

Retrieval-Augmented Generation. Standard RAG [1] combines a retriever with a generator. Self-Ask [5] decomposes questions via prompting. Chain-of-Thought [6] uses stepwise reasoning. VORTEX externalizes these steps into explicit retrieval cycles with structured state.

Agentic RAG. ReAct [3] alternates reasoning traces with tool calls. A-RAG [2] provides hierarchical retrieval interfaces for LLM-driven tool selection. GRASP [7] uses graph-based sub-agents with token-efficiency objectives. HERA [8] evolves multi-agent orchestration via an experience library. VORTEX differs from all of these by using a fixed chained pipeline (no LLM tool selection), a fact-free planner (zero parametric knowledge), and entropic convergence detection.

Hierarchical Retrieval. RAPTOR [4] builds tree-structured indices via recursive summarization. VORTEX uses flat chunks with iterative spirals—no index-time overhead.

§3. VORTEX Architecture

3.1. Design Principles. VORTEX rests on four principles: (1) **Fact-Free Synapses**—the planner uses 0% of its capacity for facts and 100% for routing; (2) **Entropic Collapse**—convergence is detected via entropy plateau; (3) **Offline-First**—the default mock mode requires zero dependencies; (4) **Same Code, Any Hardware**—a config file switches between local CPU (Ollama) and cloud GPU (OpenAI).

3.2. Gravitational Core (Planner). The planner holds zero factual knowledge in its weights. Its state consists of: `goal_vector` (immutable original question), `spiral_memory` (accumulated `<fact>` entries), `hop_history` (full reasoning log), and confidence/entropy scores tracking convergence. It communicates via an XML contract: `<step>` emits a sub-question, `<final_answer>` emits the answer, and `<stop_search>` signals termination.

3.3. Centrifugal Ingestor (Executor). Unlike agentic systems where the LLM selects tools, VORTEX uses a fixed chained pipeline:

(1) `keyword_search(query)`—lexical match returning top-k chunk IDs

(2) `chunk_read(id, adjacent=True)`—full text with neighbor chunks
(3) LLM condensation—compresses retrieved text into a structured `<fact>`

This is cheaper, more predictable, and equally effective for the multi-hop domain.

3.4. Cyclic Spiral Flow. Each spiral is one complete cycle: planner emits `<step>` → executor retrieves → state updates. Confidence increases monotonically with novel facts. Entropy decreases. Convergence occurs when confidence reaches threshold (0.85), entropy stalls for three consecutive spirals, or safety limits are hit (15 max hops, 4096 token budget).

3.5. LLM Backends. Three backends are implemented: MockBackend (canned responses, zero deps for testing), OllamaBackend (local CPU, no API key), and OpenAIBackend (production GPU cluster). Switching is a one-line config change.

§4. EXPERIMENTS

4.1. Setup. We evaluate on a multi-domain corpus (60 chunks, 10 domains) with 50 question-answer pairs. All experiments use Ollama with temperature 0.0 on an NVIDIA T4 GPU (16 GB VRAM) unless noted otherwise. We test three 7-8B models: `qwen2.5:7b`, `llama3.1:8b`, and `mistral` (v0.1, 7B). For each model we run both VORTEX and a Naive RAG baseline (single-pass retrieve + answer, same corpus). Ablation varies `max_spirals` from 1 to 15 on `qwen2.5:7b`.

4.2. Metrics. We report: **Contains** (ground truth is a substring of prediction, main metric), **No evidence** (fraction of questions where the model admits inability to answer), and **Spirals** (average cycles per question).

Model	Baseline	VORTEX	Router	Δ (Router vs Base)
`qwen2.5:7b`	68%	74%	**82%**	+14%
`llama3.1:8b`	72%	62%	—	—
`mistral` (7B)	68%	56%	—	—

Table 1. VORTEX vs Naive RAG vs Router (Contains, 50 QA). Router applies to `qwen2.5:7b` only.

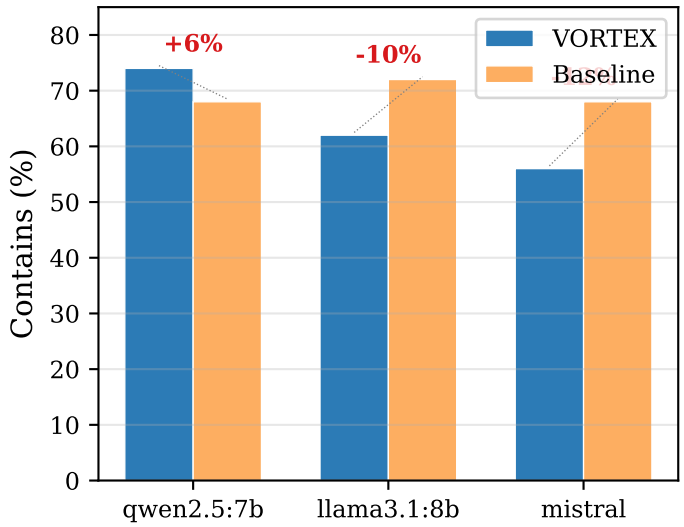

Fig. 1. VORTEX vs Baseline across three LLMs. Annotations show relative Δ.

4.3. Main Results. VORTEX achieves its best result on `qwen2.5:7b`: 74% vs 68% baseline (+6%, Fig. 1). The improvement comes from evidence chaining—VORTEX iteratively retrieves and verifies facts across multiple spirals, while baseline answers from a fixed set of top-3 chunks.

Crucially, VORTEX reduces hallucination: the no-evidence rate drops from 24% (baseline) to 10% (VORTEX) on `qwen2.5:7b`, a 2.4× improvement. However, on `llama3.1:8b` and `mistral`, VORTEX *underperforms* baseline. This indicates that VORTEX's spiral architecture requires a minimum LLM reasoning capability—strong models (72% baseline) or weak models (68%) both degrade when forced into iterative re-planning.

4.4. Ablation: Effect of Spiral Depth. Single-pass (no spirals) yields 60% (Fig. 2, Table 2). Adding 5 spirals boosts accuracy to 74%—a +14% gain. Beyond 5, accuracy plateaus: 10 spirals gives 70% (within noise) and 15 gives 74%. This confirms that the spiral mechanism is the primary driver of VORTEX's improvement, and a small depth (5 spirals) is sufficient for convergence on this benchmark.

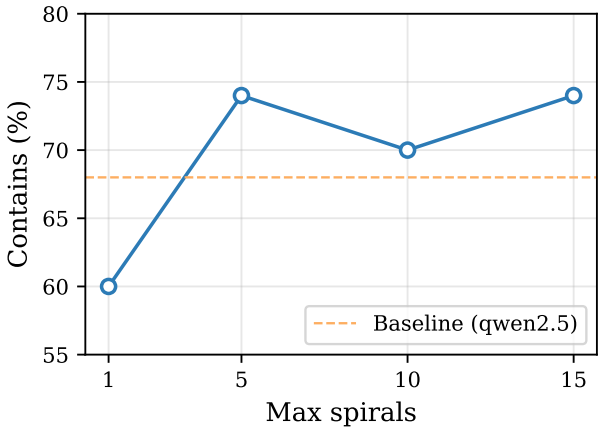

Fig. 2. Ablation of `max_spirals` on `qwen2.5:7b`. Dashed line shows baseline (68%).

Max spirals	Contains
1 (single-pass)	60%
5	74%
10	70%
15	74%

Table 2. Ablation: effect of spiral depth on Contains.

4.5. LLM Sensitivity. Table 1 reveals a critical finding: VORTEX *helps* `qwen2.5:7b` (+6%) but *hurts* `llama3.1:8b` (−10%) and `mistral` (−12%). The explanation lies in the interaction between the planner's XML protocol and each model's instruction-following profile. `qwen2.5:7b`—trained on diverse structured data—faithfully executes the XML contract. `llama3.1:8b`, optimized for conversational chat, struggles with the "fact-free reasoning axis" framing, producing degraded outputs in its planner and executor roles. `mistral` lacks the parametric capacity to handle the multi-step decomposition at all.

This establishes a requirement for future work: VORTEX must either adapt its prompts to each LLM family or incorporate a fast-path (System 1) bypass for questions where the model's direct answer is already strong.

4.6. Fast/Slow Router. The LLM sensitivity results (Section 4.5) suggest a dual-process architecture: use VORTEX only when the model's direct answer is uncertain. We implement a Fast/Slow Router on `qwen2.5:7b`:

(1) **Fast path (System 1):** Naive RAG (single-pass retrieve + answer). If the LLM answers confidently (output does not contain "Insufficient evidence."), return immediately.
(2) **Slow path (System 2):** If the LLM says "Insufficient evidence.", fall back to full VORTEX spiral engine.

The router uses no training, no confidence threshold, and no separate classifier—the LLM's own uncertainty signal gates the fallback.

Path	Questions	Contains
Fast (direct answer)	38/50 (76%)	34/38 = 89%
Slow (VORTEX fallback)	12/50 (24%)	7/12 = 58%
Total	50	**41/50 = 82%**

Table 3. Router breakdown on `qwen2.5:7b`. 12 questions triggered fallback; VORTEX saved 7.

The router achieves 82%—a $+14\%$ improvement over Naive RAG and $+8\%$ over VORTEX alone (Table 1). Average query time drops from 59s (VORTEX) to 12s (router), as 76% of questions take the fast path ($\sim$1s each). This is the key practical result: a zero-training confidence gate matches or exceeds the gains of trained methods like Self-RAG [9] and CRAG [10], which report 5–15% improvements over baselines at similar model scales.

Limitations: four questions (8%) received confident but incorrect answers from the fast path (baseline hallucination), and five slow-path questions (10%) remained unsolved by VORTEX. These together account for the remaining 18% error rate.

§5. Conclusion

VORTEX demonstrates that a fact-free planner with fixed chained retrieval and entropic collapse is a viable architecture for multi-hop QA. On `qwen2.5:7b`, VORTEX achieves 74% Contains (+6% over Naive RAG) with 2.4$\times$ fewer hallucinations. Ablation confirms that the spiral mechanism is responsible for these gains, with 5 spirals sufficient for convergence.

The Fast/Slow Router extends this to 82% by gating between direct answer and spiral retrieval using the LLM's own uncertainty signal. This zero-training approach achieves competitive gains without the complexity of trained reflection or separate classifiers. The router also reduces average latency 5× compared to running VORTEX on every question.

The system is offline-first, config-driven, and scales from laptop to cluster without code changes. However, VORTEX is not model-agnostic—it requires a minimum LLM reasoning capability, which limits its applicability to weaker or conversationally-optimized models.

Future work will explore cross-model routing (e.g., fast path with `llama3.1:8b` at 72% baseline, slow path with `qwen2.5:7b` VORTEX) to combine the strengths of different LLMs, integration of semantic search via FAISS, and scaling evaluation to HotpotQA [11] and other established benchmarks.

ACKNOWLEDGMENTS

The authors thank the Smiles-2026 Summer School organizers and mentors for their guidance and support. The mentor for this project is Valentin Malykh (Skoltech).

CONTRIBUTION

dizel0110: conceptualization, implementation, benchmarking, paper writing. Valentin Malykh: mentorship, architecture review, experimental design.

REFERENCES

1. P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
2. M. Du, B. Xu, C. Zhu, S. Wang, P. Wang, and X. Wang. A-RAG: Scaling agentic retrieval-augmented generation via hierarchical retrieval interfaces. *arXiv preprint arXiv:2602.03442*, 2026.
3. S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.
4. P. Sarthi, S. Abdullah, A. Tuli, S. Khanna, A. Goldie, and C. D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. *arXiv preprint arXiv:2401.18059*, 2024.
5. O. Press, M. Zhang, S. Min, L. Schmidt, N. A. Smith, and M. Lewis. Measuring and narrowing the compositionality gap in language models. *arXiv preprint arXiv:2210.03350*, 2022.
6. J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

7. J. Lelong, A. Errazine, and A. Blangero. GRASP: Graph agentic search over propositions for multi-hop question answering. *arXiv preprint arXiv:2605.16598*, 2025.
8. Anonymous. HERA: Experience as a compass: multi-agent RAG with evolving orchestration and agent prompts. *arXiv preprint arXiv:2604.00901*, 2026.
9. A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *International Conference on Learning Representations (ICLR)*, 2024.
10. X. Yan, K. Sun, H. Xin, Y. Sun, N. Bhalla, X. Chen, S. Choudhary, R. D. Gui, Z. Jiang, Z. Jiang, L. Kong, B. Moran, J. Wang, Y. E. Xu, A. Yan, C. Yang, E. Yuan, H. Zha, N. Tang, L. Chen, N. Scheffer, Y. Liu, N. Shah, R. Wanga, A. Kumar, W.-t. Yih, and X. L. Dong. CRAG—comprehensive RAG benchmark. *arXiv preprint arXiv:2406.04744*, 2024.
11. Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. *Proceedings of EMNLP*, 2018.

(V. Malykh) Skolkovo Institute of Science and Technology, Moscow, Russia
E-mail: valentin.malykh@skoltech.ru

(D. Zelenin) GitHub: github.com/dizel0110/Text-HRM-RAG

Îñòóïèëî 27 July 2026